Damien Robert

Deep Learning for Remote Sensing & Environment

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Postdoctoral researcher in the EcoVision lab at University of Zurich, collaborating with Jan D. Wegner. I am broadly interested in deep learning for real-world data and impactful applications, with a taste for approaches making deep learning research socially and environmentally beneficial, accessible and reproducible. My recent work focuses on efficient and scalable deep learning, forest structure analysis, and species distribution modeling.

POSITIONS

2024–now
Postdoctoral Researcher
EcoVision lab, University of Zurich / Zurich, Switzerland
Deep learning for remote sensing and environment
PI: Jan D. Wegner

2020–2024
PhD Student
ENGIE Lab CRIGEN — LASTIG, IGN/ENSG / Paris, France
Efficient learning on large-scale 3D point clouds
Advisors: Loïc Landrieu, Bruno Vallet

2017–2020
R&D Engineer
SIRADEL, ENGIE / Rennes, France
Deep learning on large-scale terrestrial/aerial, indoor/outdoor 3D and 2D data

2017
Co-Founder
Inspirama
Website gathering book recommendations from inspiring people

2015
R&D Intern
Dassault Systèmes
Dimensionality reduction and dynamic system modeling

EDUCATION

2011–2015
MSc, Engineering
École Centrale de Lyon / Lyon, France
Mathematics, Computer Science, Mechanics, Signal Processing, Automation

2022
International Computer Vision Summer School
ICVSS / Sicily, Italy
Computer vision courses by world-renowned experts in academia and industry

2017
CNRS AI Fall School
CNRS / Lyon, France
Multi-disciplinary course for AI students and researchers

AWARDS

2026
Outstanding Reviewer / CVPR

2025
PhD Award — accessit / AFRIF
French Association for Pattern Recognition and Interpretation

2024
Outstanding Reviewer / ECCV

2022
Best Paper Finalist / CVPR
Top 0.4% of submissions (33 of 8161)

PUBLICATIONS

2026 **Bias Aware Benchmark for Species Distribution Modeling**
E. Arens, N. van Tiel, R. Zbinden, C. Vanalli, **D. Robert**, L. Drees, B. Kellenberger, L. Pellissier, J. D. Wegner, D. Tuia
World Biodiversity Forum

Opportunities and challenges for near-term ecological forecasting with biodiversity and EO data
P. Singh, B. Bektas, S. Evers, E. Fenollosa, G. Koren, S. K. M. Parvathi, S. Pang, M. Paniw, **D. Robert**, G. Sialelli, J. Slingsby, R. Thornley, E. L. Underwood, J. D. Wegner, W. Wu
World Biodiversity Forum

LitePT: Lighter Yet Stronger Point Transformer Highlight
Y. Yue, **D. Robert**, J. Wang, S. Hong, J. D. Wegner, C. Rupprecht, K. Schindler
CVPR · Paper · Project · Code

EZ-SP: Fast and Lightweight Superpoint-Based 3D Segmentation
L. Geist, L. Landrieu, **D. Robert**
ICRA · Paper · Project · Code

FORMSpoT: Revealing fine-scale forest disturbances from nation-wide 1.5 m forest canopy height time series
M. Schwartz, F. Fogel, N. Besic, **D. Robert**, L. Geist, J. Renaud, J. Monnet, C. Mosig, C. Vega, A. d'Aspremont, L. Landrieu, P. Ciais
Remote Sensing of Environment · Paper

2025 **MT-GSR4B: Multi-Temporally Guided Super-Resolution for Above-Ground Biomass Estimation**
K. Karaman, V. S. F. Garnot, **D. Robert**, M. J. Santos, J. D. Wegner
PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science · Paper · Code

SSL4Eco: A Global Seasonal Dataset for Geospatial Foundation Models in Ecology
E. Plekhanova, **D. Robert**, J. Dollinger, E. Arens, P. Brun, J. D. Wegner, N. Zimmermann
CVPR workshop EarthVision · Paper · Project · Code

GSR4B: Biomass Map Super-Resolution with Sentinel-1/2 Guidance Oral
K. Karaman, Y. Jiang, **D. Robert**, V. S. F. Garnot, M. J. Santos, J. D. Wegner
ISPRS Geospatial Week · Paper · Project · Code

Climplicit: Climatic Implicit Embeddings for Global Ecological Tasks
J. Dollinger, **D. Robert**, E. Plekhanova, L. Drees, J. D. Wegner
ICLR workshop Tackling Climate Change with Machine Learning · Paper · Project · Code

Lossy Neural Compression for Geospatial Analytics: A Review
C. Gomes, I. Wittmann, **D. Robert**, J. Jakubik, T. Reichelt, M. Martone, S. Maurogiovanni, R. Vinge, J. Hurst, E. Scheurer, R. Sedona, T. Brunschwiler, S. Kesselheim, M. Batic, P. Stier, J. D. Wegner, G. Cavallaro, E. Pebesma, M. Marszalek, M. A. B. Plomer, K. Adriko, P. Fraccaro, R. Kienzler, R. Briq, S. Benassou, M. Lazzarini, C. M. Albrecht
IEEE Geoscience and Remote Sensing Magazine · Paper

2024 **Efficient Learning on Large-Scale 3D Point Clouds** AFRIF PhD Award accessit
D. Robert
PhD Thesis, Université Gustave Eiffel
Jury: Sébastien Lefèvre, Cédric Demonceaux, Patrick Pérez, Siyu Tang, Duygu Ceylan, Loic Landrieu, Bruno Vallet · Paper · Code

Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering Oral
D. Robert, H. Raguet, L. Landrieu
3DV · top 5.3% of submissions · Paper · Project · Code

2023 **Efficient 3D Semantic Segmentation with Superpoint Transformer**
D. Robert, H. Raguet, L. Landrieu
ICCV · top 26.8% of submissions · Paper · Project · Code

2022 **Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation** Oral and Best paper finalist
D. Robert, B. Vallet, L. Landrieu
CVPR · top 0.4% of submissions · Paper · Project · Code

UNDER REVIEW

2026	**Toward a Foundation Model for Forest Point Clouds** Y. Yue, S. Puliti, **D. Robert**, A. Topaloğlu, B. Xiang, M. Wielgosz, J. D. Wegner, R. Astrup, C. Rupprecht, K. Schindler
2026	**From Machine Learning to Large-Scale EO Products: Best Practices for Making Maps** G. Sialelli, R. Young, Y. Jiang, C. Aybar, L. Scheibenreif, **D. Robert**, C. Mosig, A. J. Stewart, J. D. Wegner, A. Pirinen, O. Mogren, K. Schindler

COLLABORATION & SUPERVISION

PhD Students	**Emilia Arens** UZH with Jan D. Wegner · **Johannes Dollinger** UZH with Jan D. Wegner · **Kaan Karaman** UZH with Jan D. Wegner · **Louis Geist** ENPC with Loïc Landrieu · **Yuanwen Yue** ETH / Oxford with Konrad Schindler and Christian Rupprecht
Master Students	**Valerio Schelbert** ETH · **Jackson Sunny** École Polytechnique with SAMP
Collaborators	**Konrad Schindler** ETH · **Christian Rupprecht** Oxford · **Jan D. Wegner** UZH · **Loïc Landrieu** ENPC · **Vivien Sainte Fare Garnot** UZH · **Lukas Drees** UZH · **Martin Schwartz** LSCE · **Elena Plekhanova** WSL
Advising	**UpCircle AI** Switzerland · **École des Sciences Criminelles de Lausanne** Switzerland · **SAMP AI** France · **GEODIGITAL** US

REVIEWING & CHAIRING

2026	ECCV Area Chair · CVPR — Outstanding Reviewer
2025	CVPR · CVPR workshop EarthVision
2024	ECCV — Outstanding Reviewer · ISPRS Journal of Photogrammetry and Remote Sensing · ICLR workshop ML4RS · CVPR workshop EarthVision
2023	CVPR · CVPR workshop EarthVision
2022	ISPRS Journal of Photogrammetry and Remote Sensing · CVPR workshop EarthVision

TEACHING

2026	**Transformer** / UZH · Lecture, Master 2	
2025	**AI4Good group projects** / UZH · Labs, Master 2	14 h
2025	**Transformer** / UZH · Lecture, Master 2	4 h
2024	**Transformer, NeRFs, and Diffusion** / UZH · Lecture and labs, Master 2	5.5 h
2023	**Deep Learning for Remote Sensing** / ENSG-IGN · Lecture, Master 2	13 h
2022	**3D Deep Learning for Remote Sensing** / ISPRS'22 · Tutorial, Researchers	1 day
2022	**3D Deep Learning, Torch-Points3D & DeepViewAgg** / ENGIE CRIGEN lab · Tutorial, Researchers	1 day
2022	**Deep Learning for Remote Sensing** / ENSG-IGN · Lecture, Master 2	9 h
2020	**Deep Learning for Computer Vision** / Ecole Polytechnique · Lecture, Master 1	12 h

OPEN-SOURCE SOFTWARE

★ 405	**LitePT** / prs-eth/LitePT	44 forks
★ 15	**torch-graph-components** / drprojects/torch-graph-components	
★ 1054	**Superpoint Transformer** / drprojects/superpoint_transformer	136 forks
★ 53	**NoRA** / drprojects/nora	3 forks
★ 74	**Point Geometric Features** / drprojects/point_geometric_features	8 forks
★ 237	**DeepViewAgg** / drprojects/DeepViewAgg	24 forks

SELECTED TALKS

2026 **ESA-NASA workshop on AI Foundation Model for EO** / tutorial · Building Earth Observation Embedding Workflows with TerraTorch · Virtual

ESA Φ-lab · Geospatial foundation models for forest disturbance monitoring · Frascati, Italy

2025 **MPI of Animal Behavior** · Research at EcoVision lab · Zurich, Switzerland

Global Forest Watch · Geospatial foundation models for forest disturbance monitoring · Virtual

NatureFinance · Research at EcoVision lab · Virtual

NVIDIA Strategic Research Engagement · Superpoint Transformer and EcoVision research · Zurich, Switzerland

CNES COMET TSI · Remote Sensing for Species Distribution Modeling · Virtual

2024 **3D Data Academy** / interview · Interview by Florent Poux and tutorial for Superpoint Transformer · YouTube

Google DeepMind · Research at EcoVision lab · Zurich, Switzerland

3DV'24 / oral · Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering · Davos, Switzerland

2023 **Ecole des Ponts, IMAGINE lab** · Efficient Learning on Large-Scale 3D Point Clouds · Paris, France

National Land Survey of Finland (NLS) · IGN's research on Large-Scale 2D and 3D Learning · Paris, France

ETH Zürich, Computer Vision and Geometry lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland

ETH Zürich, Photogrammetry and Remote Sensing lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland

Valeo.ai · Efficient 3D Semantic Segmentation with Superpoint Transformer · Paris, France

2022 **CV News** / interview · Interviewed by the CV News journal for its Best of CVPR'22 issue · New Orleans, US

CVPR'22 / oral · Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation · New Orleans, US

45 talks given since 2021 — full list at drprojects.github.io/talks.

SKILLS & LANGUAGES

Research **3D & geometry** Large-scale 3D point clouds · LiDAR data · Superpoint-based learning · Graph-based learning and graph partitioning

Efficient & scalable learning Efficient deep learning · Scalable model design · EO data representation and compression

Learning paradigms Multimodal learning · Self-supervised learning · Representation learning

Environment & Earth observation Remote sensing · Forest structure analysis · Species distribution modeling · Representation learning for macroecology

Tools Python · PyTorch · PyTorch Lightning · PyTorch Geometric · C++ · Hydra · Weights & Biases · Git · Linux · SLURM · scikit-learn · Plotly · Blender · LaTeX

Languages French Native · English Fluent · Spanish Intermediate · German Beginner

Abroad Zurich, Switzerland 2024–now · Backpacking, Worldwide 2015–2016 · Providence, RI, United States 2014–2015

Interests Nature & forests · Boxing · Mountaineering · Skateboarding · Guitar · Painting · Reading · Board games